Assessment 2

MIS605 Systems Analysis and Design - Design Specification
Case study: Doing Good Rewards (DGR)

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Learning Facilitator Name:
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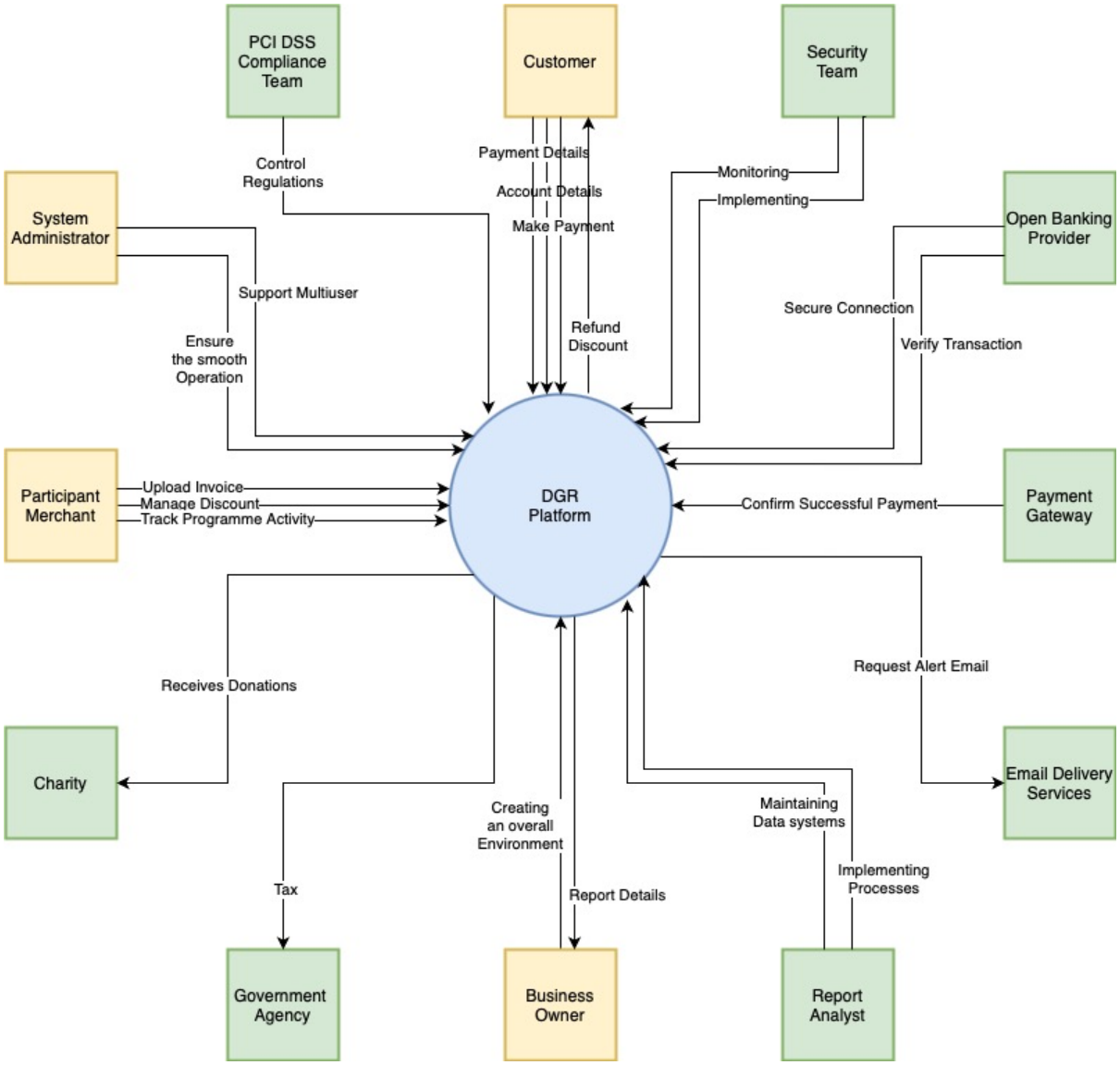
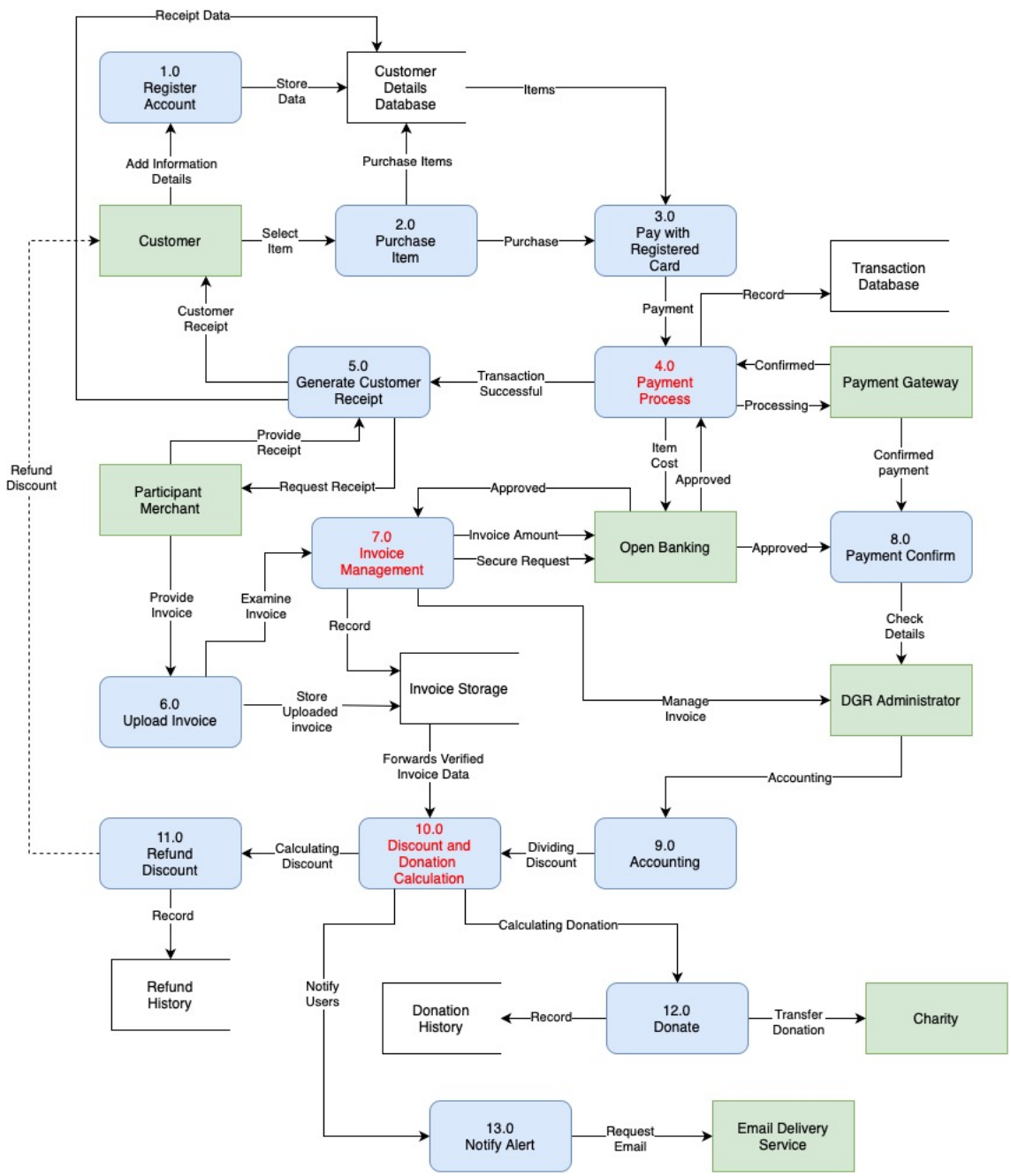
Question 1 (15 mark).

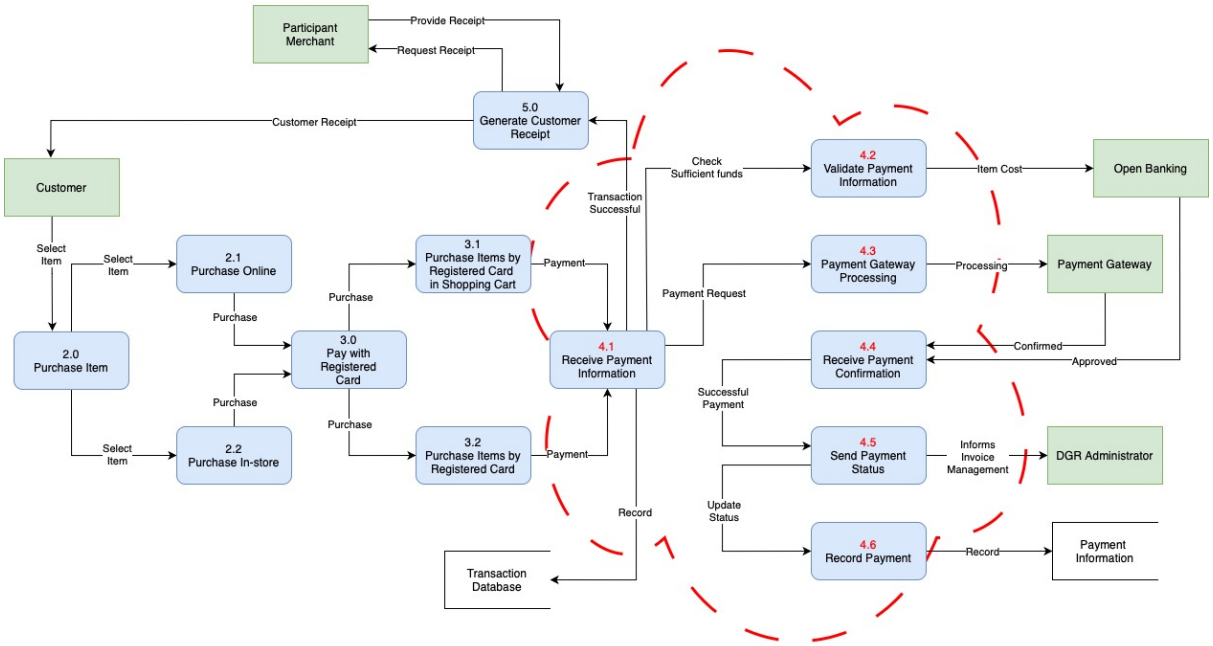


Question 2 (20 mark).

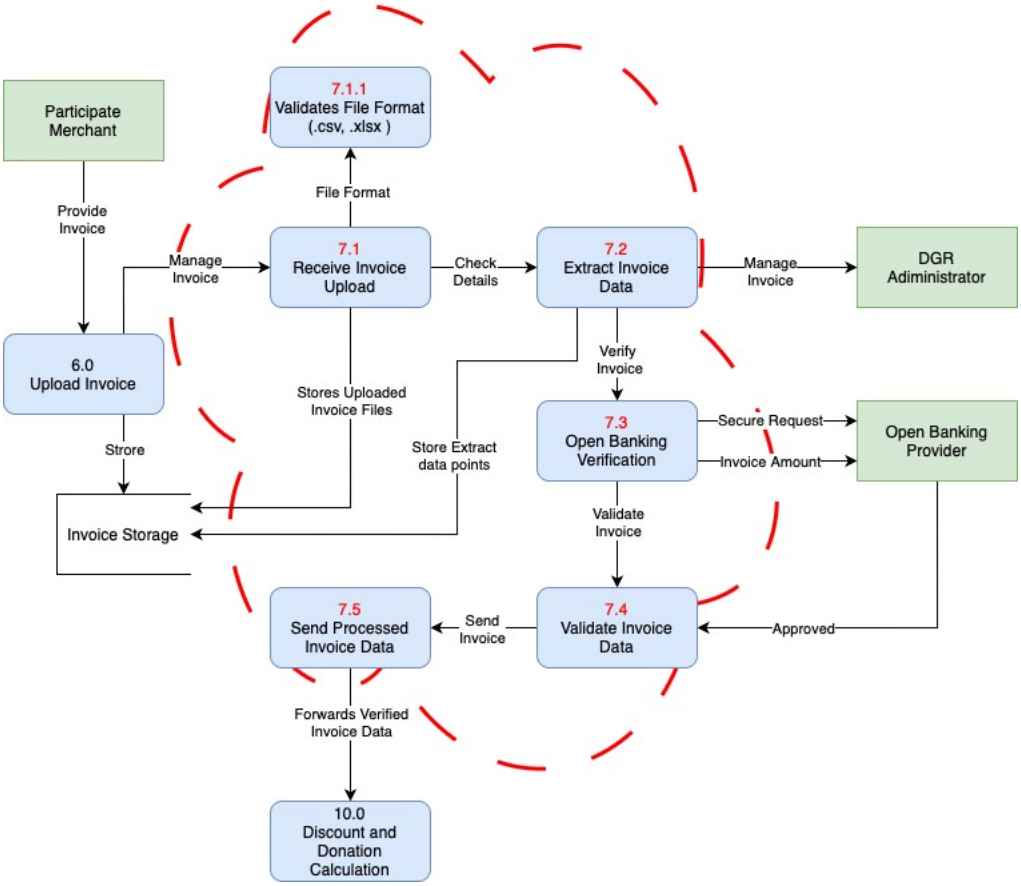


Question 3 (24 mark).

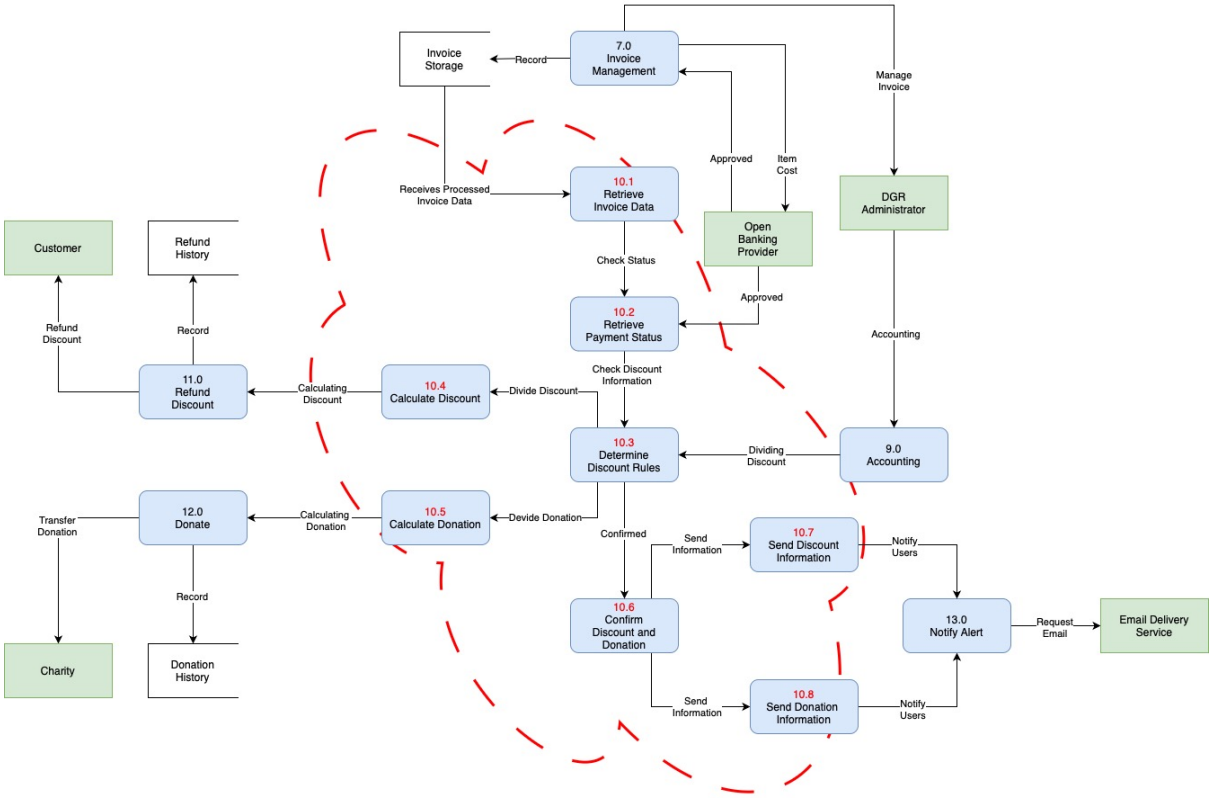
4.0 Payment Processing



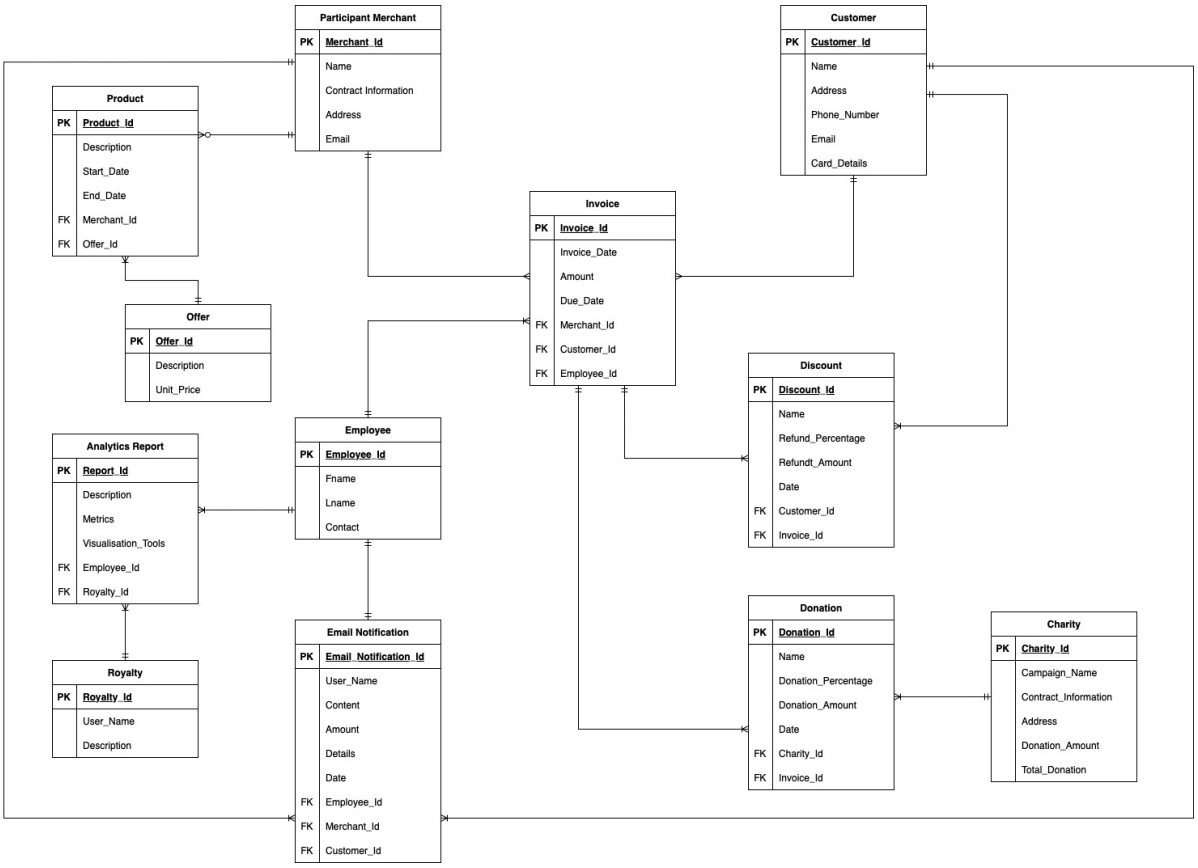
7.0 Invoice Management



10.0 Discount and Donation Calculation



Question 4 (20 mark).



Question 5 (5 mark).

A relational database (RDBMS) is a type of database that stores data in tables. Each table represents a specific entity (e.g., customers, products) or relationship between entities. Tables are made up of rows (records) and columns (attributes).

Key Characteristics:

- Tabular data: Data is organized in a clear and easy-to-understand table format.
- Relationships: Tables can be linked using keys (primary and foreign), enabling connections between related data.
- Data integrity: Rules are enforced to ensure data consistency and accuracy.
- SQL: Structured Query Language (SQL) is used to interact with the database (query, update, insert, delete data).

Advantages:

- Simplicity: Easy to understand and use, even for non-technical users.
- Widely adopted: Popular choice with a vast ecosystem of tools, software, and expertise.
- Data integrity: Ensures data consistency and reduces errors.
- Scalability: Can handle large data volumes and high transaction rates.

Common Use Cases:

- Transactional applications (e.g., e-commerce, banking): Frequent data updates (inserts, updates, deletes).
- Data warehousing: Stores vast data volumes for analysis and business intelligence.
- Master data management: Manages core information (customers, products, employees).

Why Choose RDBMS?

- Structured Data: If the case study deals with well-defined entities and their attributes (e.g., donors, donations, projects), an RDBMS excels at storing such structured data in tables with rows and columns.
- Relationships: Entities likely have relationships (donors contribute to projects). RDBMS excels at modelling these connections through foreign keys.
- Data Integrity: RDBMS ensures data consistency through constraints and data types, crucial for managing donor information and project data accurately.
- Querying and Reporting: Generating reports on donations, analysing project progress, and managing data requires efficient querying capabilities. RDBMS provides powerful tools for such tasks.

Additional Considerations:

- Data Volume and Complexity: If the case study involves a massive amount of data or complex relationships, explore advanced RDBMS features, or consider alternative data stores for specific data types (e.g., unstructured data).
- Scalability: If the DGR organization anticipates significant data growth, an RDBMS that scales efficiently is essential.
- Security: RDBMS with robust security measures is crucial for protecting sensitive donor information and project data.

By evaluating these factors and the specific data management needs outlined in the DGR case study, you can determine if an RDBMS is the optimal data storage solution.

Question 6 (16 mark).

1. **Participate Merchants Table:**

Original Entity:

Participant Merchant	
PK	Merchant_Id
	Name
	Contract Information
	Address
	Email

Tables in 3NF:

Merchants Table

MerchantID	Name
M001	Acme Electronics
M002	Green Goods Co.
M003	Best Buy
M004	Tech Toys
M005	Sports Warehouse

Merchant Details Table

MerchantDetailID	MerchantID	ContractInformation	Address	Email
MD001	M001	Phone…, 10-5pm	123 Main St, Anytown, NSW 12345	[email]
MD002	M002	Phone…, open8am	456 Elm St, Anytown, NSW 12345	[email]
MD003	M003	Phone…, 8-3pm	789 Oak St, Anytown, NSW 12345	[email]
MD004	M004	Phone…, 8-4pm	101 Pine St, Anytown, NSW 12345	[email]
MD005	M005	Phone…, 10-6pm	202 Maple St, Anytown, NSW 12345	[email]

2. Customer

Original Entity:

Customer	
PK	Customer_Id
	Name
	Address
	Phone_Number
	Email
	Card_Details

Tables in 3NF:

Customers Table

CustomerID	Name
C001	John Doe
C002	Jane Smith
C003	Michael Brown
C004	Alice Johnson
C005	David Williams

Customer Details Table

CustomerDetailID	CustomerID	Address	Phone_Number	Email
CD001	C001	123 Main St, Anytown, NSW 12345	+61 (55) 555-1234	[email]
CD002	C002	456 Elm St, Anytown, NSW 12345	+61 (55) 555-5678	[email]
CD003	C003	789 Oak St, Anytown, NSW 12345	+61 (55) 555-9012	[email]
CD004	C004	101 Pine St, Anytown, NSW 12345	+61 (55) 555-2345	[email]
CD005	C005	202 Maple St, Anytown, NSW 12345	+61 (55) 555-6789	[email]

Payment Methods Table

PaymentMethodID	CustomerID	CardType	CardNumber	ExpirationDate	CVV
PM001	C001	Visa	1234567890123456	2024-12-31	123
PM002	C002	Mastercard	9876543210987654	2025-06-30	456

3. Invoice

Original Entity:

Invoice	
PK	Invoice_Id
	Invoice_Date
	Amount
	Due_Date
FK	Merchant_Id
FK	Customer_Id
FK	Employee_Id

Tables in 3NF:

Invoice Table

InvoiceID	InvoiceDate	Amount (AUD)	DueDate	MerchantID	CustomerID	EmployeeID
INV001	2024-04-10	100	2024-05-10	M001	C002	E003
INV002	2024-04-12	50	2024-04-22	M003	C003	E002
INV003	2024-04-15	25	2024-05-05	M005	C004	E003

4. **Product**

Original Entity:

Product	
PK	Product_Id
	Description
	Start_Date
	End_Date
FK	Merchant_Id
FK	Offer_Id

Tables in 3NF:

Products Table

ProductID	Name	Description
001	Widget X	A useful tool for various tasks.
002	Gadget Y	A cutting-edge device with advanced features.
003	Gizmo Z	A fun and interactive toy.

Product Details Table

ProductDetailID	ProductID	StartDate	EndDate	MerchantID	OfferID
PD001	001	2024-04-01	2024-06-30	M001	OFF005
PD002	002	2024-03-15	NULL	M005	OFF010
PD003	003	2024-04-10	2024-12-31	M013	OFF015

5. **Offer**

Original Entity:

Offer	
PK	Offer_Id
	Description
	Unit_Price

Tables in 3NF:

Offer Table

OfferID	Description	UnitPrice
OFF001	Summer Sale	Discount 5% off
OFF002	Midyear Sale	Discount 10% off to $50
OFF003	Grand Sale	Up to 15% off

6. **Employee**

Original Entity:

Employee	
PK	Employee_Id
	Fname
	Lname
	Contact

Tables in 3NF:

Employee Table

EmployeeID	FirstName	LastName	Email
E001	John	Doe	[email]
E002	Jane	Smith	[email]
E003	Michael	Brown	[email]
E004	Alice	Johnson	[email]
E005	David	Williams	[email]

7. Analytics Report

Original Entity:

Analytics Report	
PK	Report_Id
	Description
	Metrics
	Visualisation_Tools
FK	Employee_Id
FK	Royalty_Id

Tables in 3NF:

Reports Table:

ReportID	Description	Metrics	VisualizationTools
R001	Sales Report Q1 2024	Revenue, Conversion Rate, Customer Acquisition	Bar Charts, Line Graphs
R002	Marketing Campaign Analysis	Click-Through Rate, Impression Share, Engagement Metrics	Pie Charts, Heatmaps
R003	Inventory Management Report	Stock Levels, Product Performance, Reorder Points	Tables, Dashboards

Report Details Table:

ReportDetailID	ReportID	EmployeeID	RoyaltyID
RD001	R001	E001	NULL
RD002	R002	E010	RY002
RD003	R003	E003	NULL

8. Royalty

Original Entity:

Royalty	
PK	Royalty_Id
	User_Name
	Description

Tables in 3NF:

RoyaltyID	Username	Description
RY001	user_abc123	Royalty for Advance Level Donation.
RY002	creator_xyz	Royalty for Beginner Level Donation.
RY003	author_def456	Royalty for Beginner Level Donation.

9. Email Notification

Original Entity:

Email Notification	
PK	Email_Notification_Id
	User_Name
	Content
	Amount
	Details
	Date
FK	Employee_Id
FK	Merchant_Id
FK	Customer_Id

Tables in 3NF:

Notification Table

NotificationID	Content	Details	Date
N001	System Update Reminder	Please update your app to the latest version.	2024-04-15
N002	Order Confirmation	Your order #12345 has been confirmed!	2024-04-14
N003	Promotional Offer	Successful Payment, your 5% discount will refund to your account on 20 April	2024-04-13

Notification Details Table

NotificationDetailID	NotificationID	Username	Amount	EmployeeID	MerchantID	CustomerID
ND001	N001	user_abc123	10.0	E003	M002	C001
ND002	N002	customer_xyz	55.0	E003	M001	C002
ND003	N003	author_def456	30.2	E004	M004	C003

10. <u>Discounts</u>

Original Entity:

Discount	
PK	Discount_Id
	Name
	Refund_Percentage
	Refundt_Amount
	Date
FK	Customer_Id
FK	Invoice_Id

Tables in 3NF:

Discounts Table:

DiscountID	Name	Date	CustomerID	InvoiceID
D001	Spring Sale Discount	2024-04-10	C002	INV001
D002	Weekend Special	2024-04-13	C001	INV003
D003	Loyalty Reward	2024-04-12	C005	INV002

DiscountDetails Table

DiscountDetailID	DiscountID	Refund_Percentage (%)	Refund_Amount (AUD)
DD001	D001	90%	2.50
DD002	D002	80%	10.00
DD003	D003	0%	0.00

11. Donation

Original Entity:

Donation	
PK	Donation_Id
	Name
	Donation_Percentage
	Donation_Amount
	Date
FK	Charity_Id
FK	Invoice_Id

Tables in 3NF:

Donations Table:

DonationID	Name	Date	CharityID	InvoiceID
DN001	Spring Charity Drive	2024-04-10	CH001	INV020
DN002	Weekend Donation	2024-04-14	CH002	INV040
DN003	Birthday Donation	2024-04-12	CH003	INV010

DonationDetails Table

DonationDetailID	DonationID	Donation_Percentage (%)	Donation_Ampunt (AUD)
DND001	DN001	10%	0.50
DND002	DN002	20%	2.00
DND003	DN003	100%	10.00

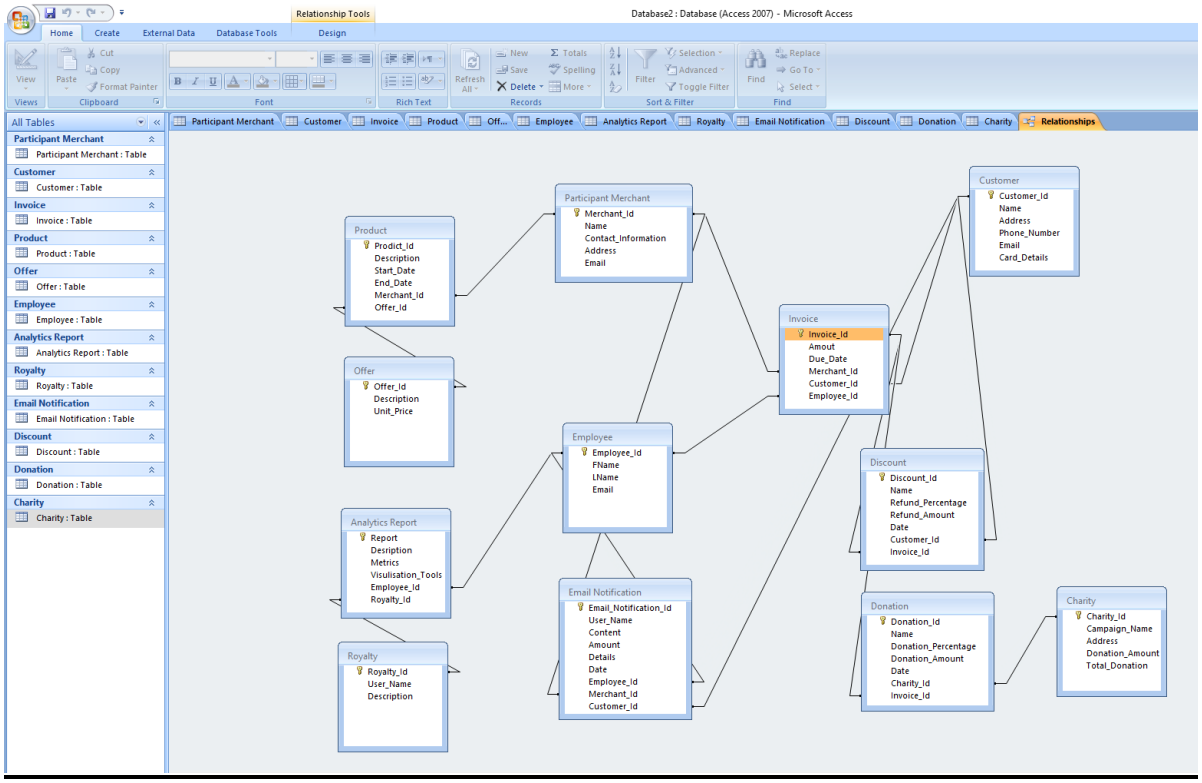
12. Charity

Original Entity:

Charity	
PK	Charity_Id
	Campaign_Name
	Contract_Information
	Address
	Donation_Amount
	Total_Donation

Tables in 3NF:

Charity Table

CharityID	Campaign_Name	ContactInfo	Address	Donation_Amount (AUD)	Total_Donation (AUD)
CH001	Hope Haven Animal Shelter	(61)123-4567	1 Main St, Anytown, NSW 12345	10.00	8,000.00
CH002	Children's Literacy Foundation	(61)789-0123	45 Elm St, Anytown, VIC 54321	5.00	10,000.00

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